



Centre for
Robotics



SLAM Course

09 - Photogrammetry

From Projective Geometry to Bundle Adjustment

Dr Alejandro Fontán Villacampa
April 2026

Course Material Attribution

These slides were created by **Alejandro Fontan** as part of the **SLAM course**.

I have made every effort to acknowledge all sources from which slides, images, or videos were used. If I have made an error or omission, please let me know.

Special acknowledgments

- ❖ **Cyrill Stachniss** — *Photogrammetry and Robotics*. These slides draw heavily on Cyrill Stachniss's course material. That material in turn relied on the lecture notes by Wolfgang Förstner and the book Photogrammetric Computer Vision by Förstner & Wrobel. Parts of the robotics content are based on Probabilistic Robotics by Thrun, Burgard, and Fox.
- ❖ **Kaveh Fathian** — *Robotic Mapping and Localization*, Colorado School of Mines, Department of Computer Science. These slides draw heavily on the course material, which was itself compiled from several courses, lectures, and code repositories. Corresponding references are cited at the end of each lecture.
- ❖ **Antonio Torralba, Phillip Isola, and William T. Freeman** — *Foundations of Computer Vision*. Some images are sourced from this book.
- ❖ **L. Carlone, A. Kim, T. Barfoot, D. Cremers, and F. Dellaert** — *SLAM Handbook: From Localization and Mapping to Spatial Intelligence*, eds.. Cambridge University Press, 2026.

For Educators: university lecturers are welcome to use this material freely. If you adapt it, please retain all existing acknowledgments and add an acknowledgment to **Alejandro Fontan**. As a personal favour, I would appreciate an email notice if you use these slides.

Alejandro Fontan: alejandro.fontan@qut.edu.au

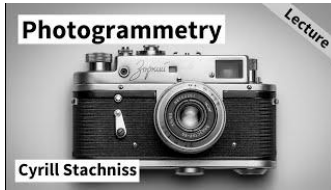


Photogrammetry



5 Minutes with Cyrill

<https://www.youtube.com/playlist?list=PLgnQpQtFTOGSO8HC48K9sPuNliY1qzxV9>



Photogrammetry I & II

<https://www.youtube.com/playlist?list=PLgnQpQtFTOGRYjqjDZxTEQPZuFHQa707Y>



SLAM Handbook

<https://github.com/SLAM-Handbook-contributors/slam-handbook-public-release>

Foundations of Computer Vision

<https://visionbook.mit.edu/>

L. Carlone, A. Kim, T. Barfoot, D. Cremers, and F. Dellaert, *SLAM Handbook: From Localization and Mapping to Spatial Intelligence*, Eds. Cambridge University Press, 2026.

A. Torralba, P. Isola, and W.T. Freeman, *Foundations of Computer Vision*, Eds. Cambridge University Press, 2026.



<https://github.com/alejandrofontan/SLAM-Course/>
<https://github.com/ariarobotics/robotic-mapping>



<https://github.com/VSLAM-LAB/VSLAM-LAB>



Geometry and Images



Geometry and Images



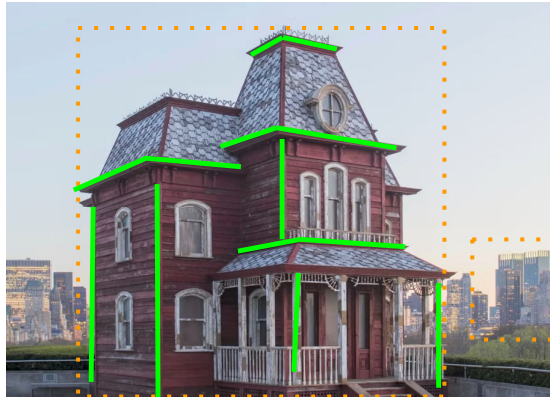
Geometry and Images

- ❖ **Not length-preserving:** size of objects is inverse proportional to the distance



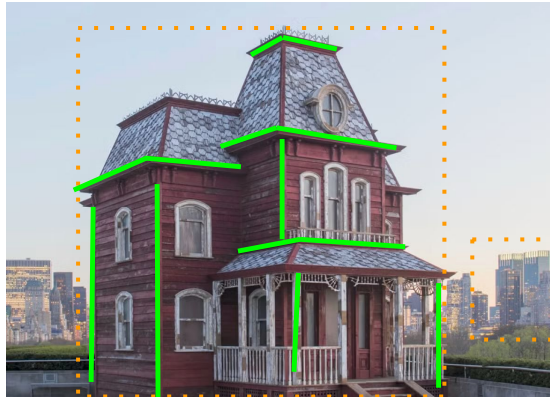
Geometry and Images

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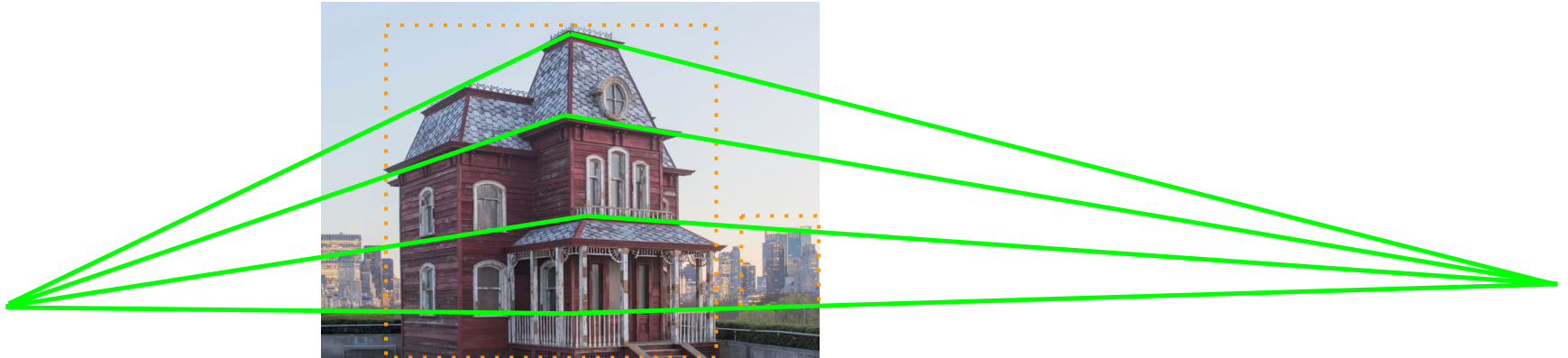
Geometry and Images

- ❖ **Not length-preserving**: size of objects is inverse proportional to the distance
- ❖ **Not angle-preserving**: angles between lines change
- ❖ **Line-preserving**: straight lines are mapped to straight lines



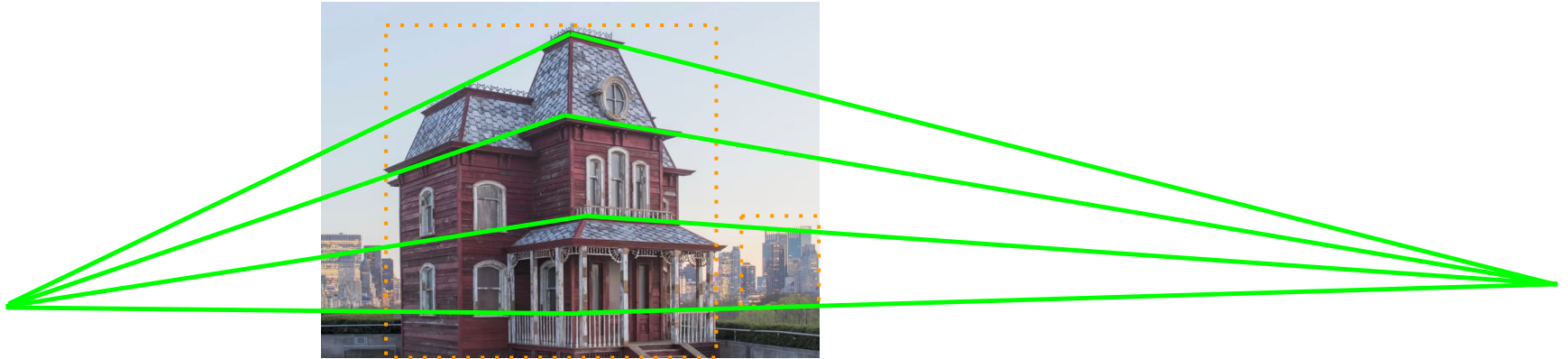
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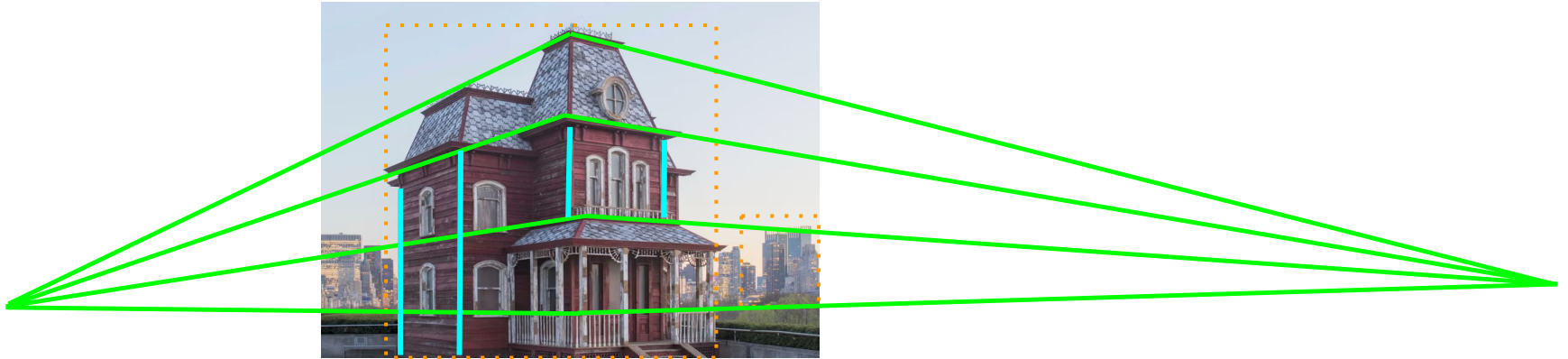
Geometry and Images

- ❖ **Not length-preserving**: size of objects is inverse proportional to the distance
- ❖ **Not angle-preserving**: angles between lines change
- ❖ **Line-preserving**: straight lines are mapped to straight lines
- ❖ **Parallel lines may not remain parallel**, all intersect in a vanishing point
 - The **vanishing point** is the “point at infinity” for the parallel lines
 - Every direction has exactly one vanishing point



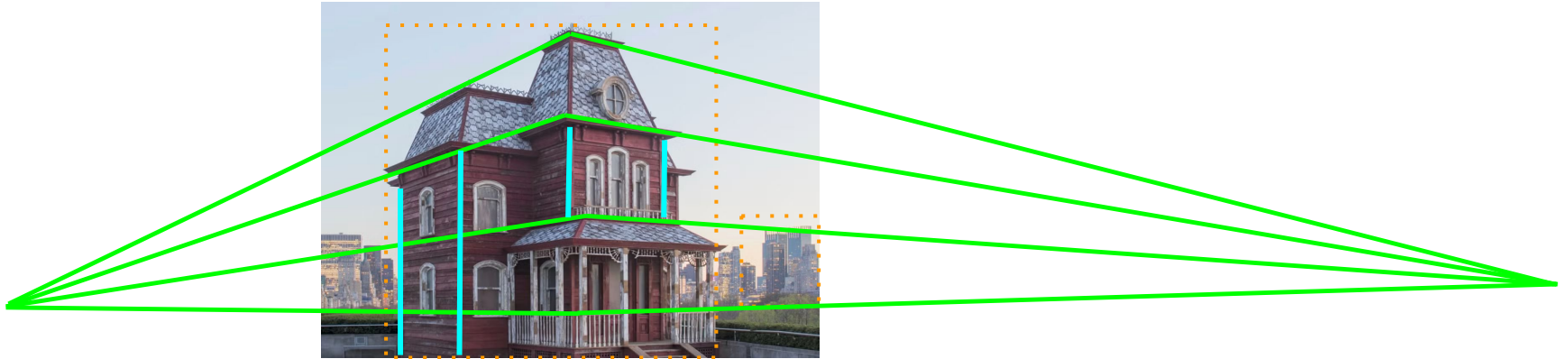
Geometry and Images

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Geometry and Images

- ❖ **Not length-preserving**: size of objects is inverse proportional to the distance
- ❖ **Not angle-preserving**: angles between lines change
- ❖ **Line-preserving**: straight lines are mapped to straight lines
- ❖ **Parallel lines may not remain parallel**, all intersect in a vanishing point
 - The **vanishing point** is the “point at infinity” for the parallel lines
 - Every direction has exactly one vanishing point
 - Parallel lines that are also parallel to the image plane remain parallel in the projection (they meet at a vanishing point “at infinity,” which is never visible)



Euclidian Geometry vs Projective Geometry

Euclidian Geometry

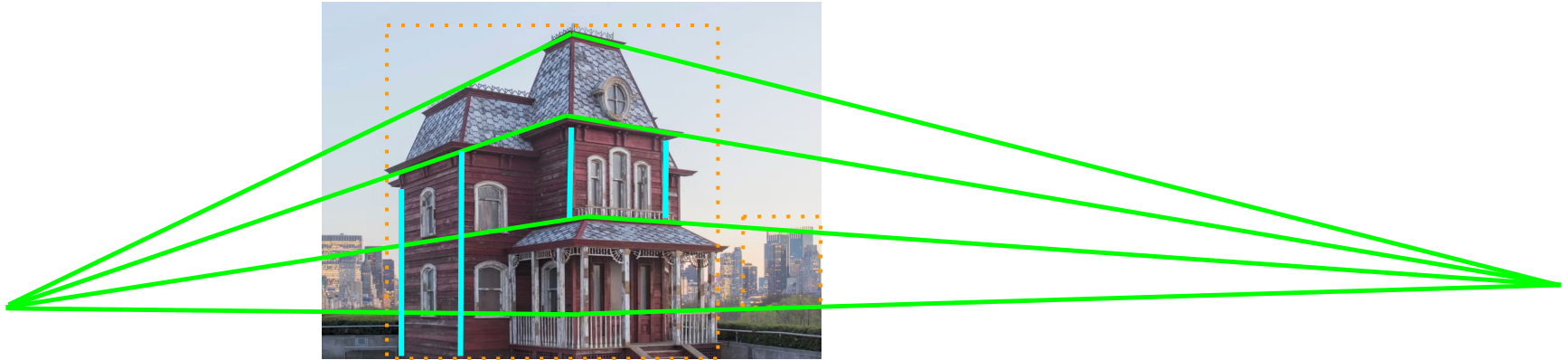
Is suboptimal to describe the central projection

Projective Geometry

Is an alternative algebraic representation of geometric objects and transformations

Homogeneous Coordinates

- 1) Points at infinity can be represented using finite coordinates
- 2) A single matrix can represent affine and projective transformations



Homogeneous Coordinates

Definition: The representation $\mathbf{x}$ of a geometric object is **homogeneous** if $\mathbf{x}$ and $\lambda\mathbf{x}$ represent the same object for $\lambda \neq 0$.

Euclidian Coordinates

$$\mathbf{x} \neq \lambda\mathbf{x}$$

Euclidian Coordinates of a point in the plane is a **2-dim. vector**

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Example for $\mathbb{R}^2 \rightarrow \mathbb{P}^2$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 2 \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2 \end{bmatrix} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix}$$

Homogeneous Coordinates

$$\mathbf{x} = \lambda\mathbf{x}$$

Homogeneous Coordinates of a point in the plane is a **3-dim. vector**

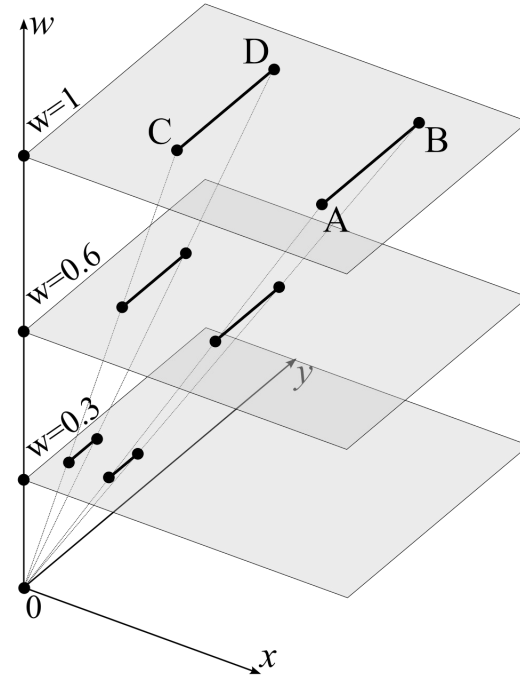
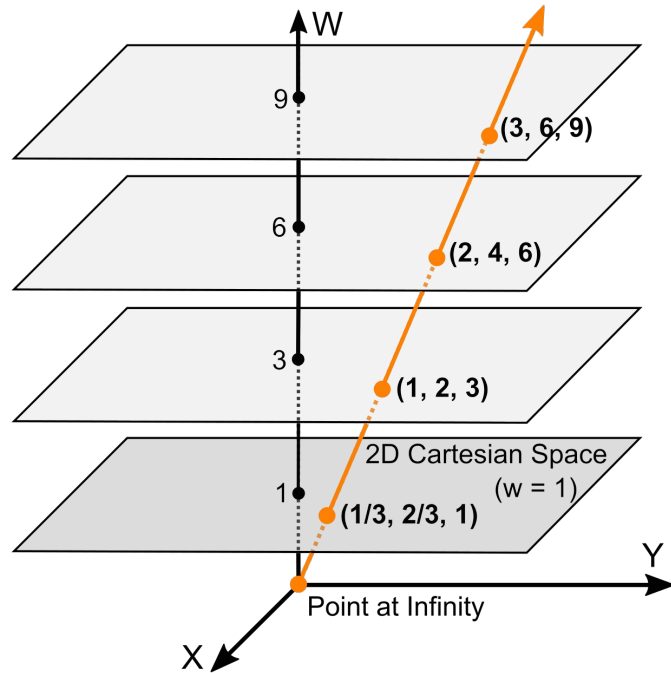
$$\chi: \quad \mathbf{x} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad \text{with } |\mathbf{x}|^2 = u^2 + v^2 + w^2 \neq 0$$

Example for $\mathbb{P}^2 \rightarrow \mathbb{R}^2$

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} u/w \\ v/w \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} u/w \\ v/w \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$



Points at Infinity

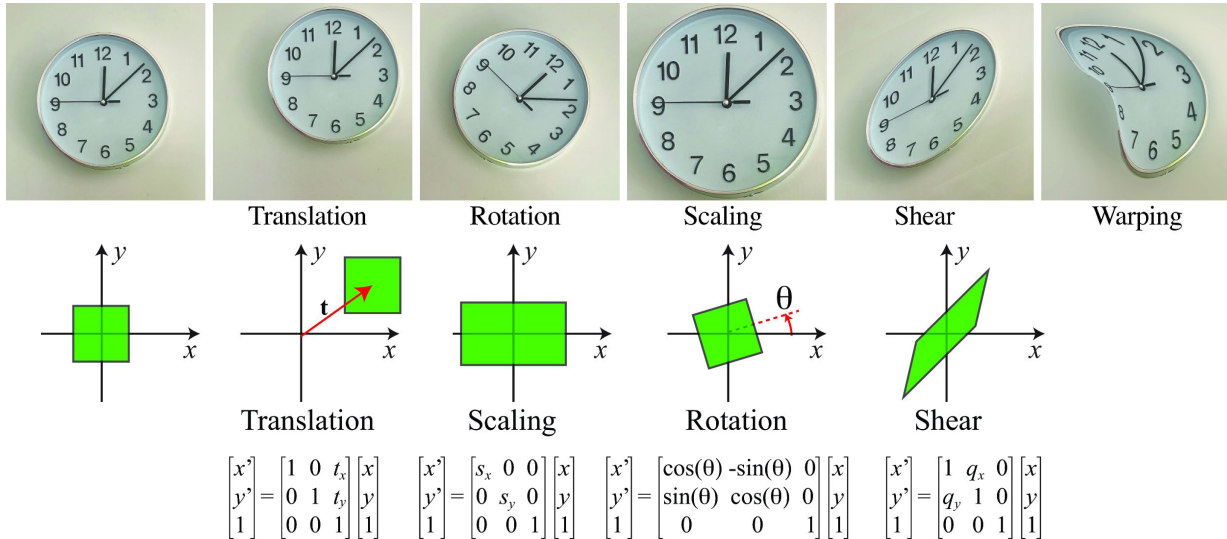


Transformations

A projective transformation is an invertible linear mapping

$$\mathbf{x}' = H\mathbf{x}$$

How should H look like to realize relevant transformation?

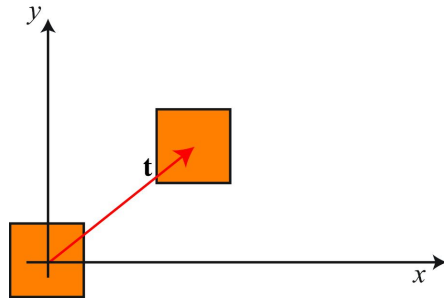


Transformations: Translation

Heterogeneous Coordinates

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

Homogeneous Coordinates



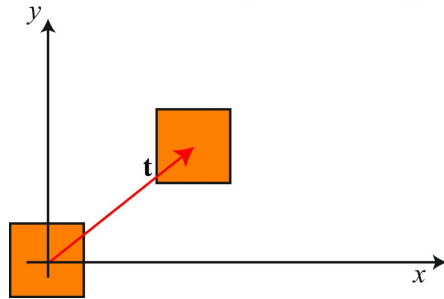
Transformations: Translation

Heterogeneous Coordinates

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

Homogeneous Coordinates

$$\mathbf{p}' = \mathbf{T}\mathbf{p}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



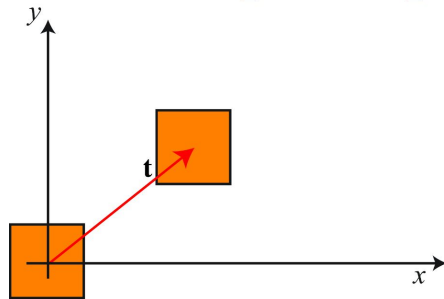
Transformations: Translation

Heterogeneous Coordinates

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Homogeneous Coordinates

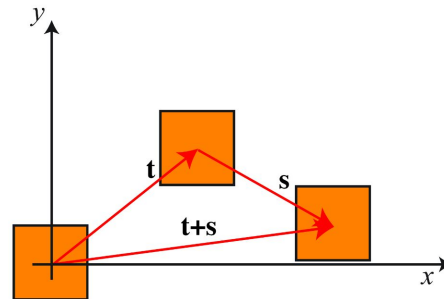
$$\mathbf{p}' = \mathbf{T}\mathbf{p}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



Heterogeneous Coordinates

$$\mathbf{p}' = \mathbf{p} + \mathbf{t} + \mathbf{s}$$

Homogeneous Coordinates



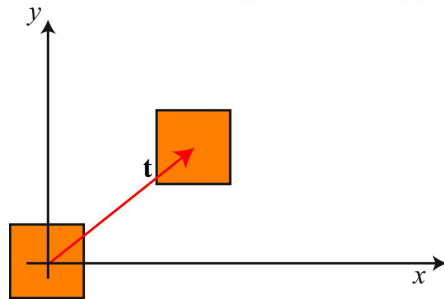
Transformations: Translation

Heterogeneous Coordinates

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

Homogeneous Coordinates

$$\mathbf{p}' = \mathbf{T}\mathbf{p}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

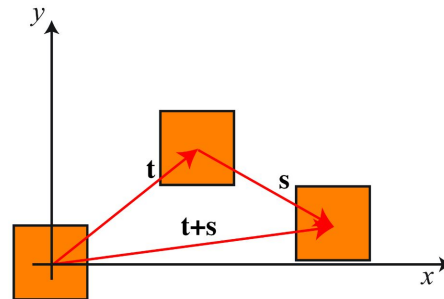


Heterogeneous Coordinates

$$\mathbf{p}' = \mathbf{p} + \mathbf{t} + \mathbf{s}$$

Homogeneous Coordinates

$$\mathbf{p}' = \mathbf{S}\mathbf{T}\mathbf{p}, \quad \mathbf{S} = \begin{bmatrix} 1 & 0 & s_x \\ 0 & 1 & s_y \\ 0 & 0 & 1 \end{bmatrix}$$



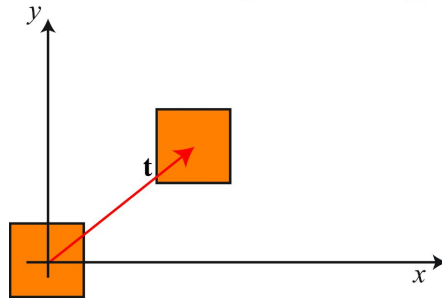
Transformations: Translation

Heterogeneous Coordinates

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

Homogeneous Coordinates

$$\mathbf{p}' = \mathbf{T}\mathbf{p}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

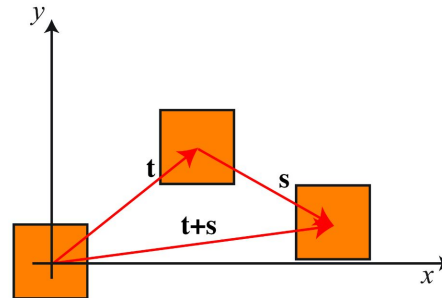


Heterogeneous Coordinates

$$\mathbf{p}' = \mathbf{p} + \mathbf{t} + \mathbf{s}$$

Homogeneous Coordinates

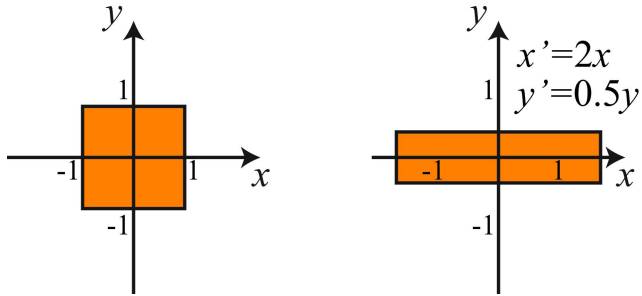
$$\mathbf{p}' = \mathbf{S}\mathbf{T}\mathbf{p}, \quad \mathbf{S} = \begin{bmatrix} 1 & 0 & s_x \\ 0 & 1 & s_y \\ 0 & 0 & 1 \end{bmatrix}$$



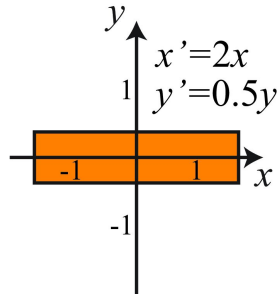
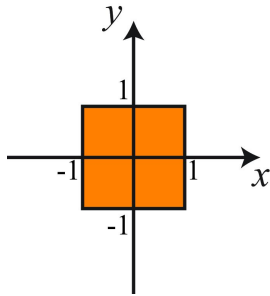
$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & s_x \\ 0 & 1 & s_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x + s_x \\ 0 & 1 & t_y + s_y \\ 0 & 0 & 1 \end{bmatrix}$$



Transformations: Scaling



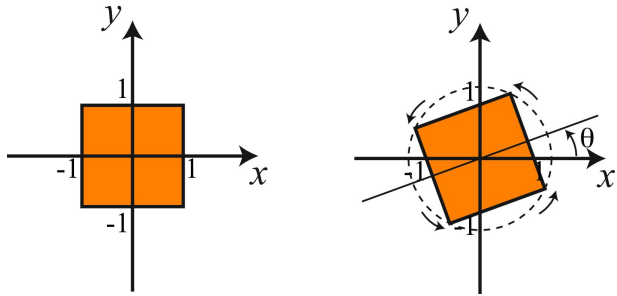
Transformations: Scaling



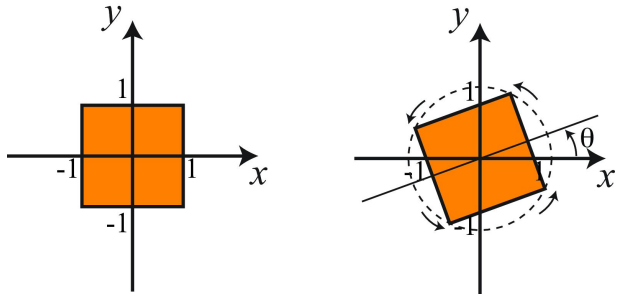
$$\mathbf{S} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Transformations: Rotation



Transformations: Rotation

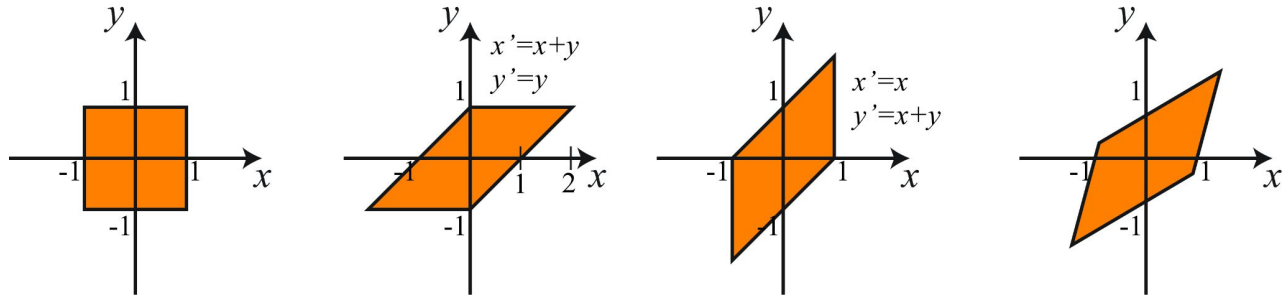


$$\mathbf{p}' = \mathbf{R}\mathbf{p},$$

$$\mathbf{R} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Transformations: Shearing



$$Q = \begin{bmatrix} 1 & q_x & 0 \\ q_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



3D Transformations

Translation (3 parameters)

$$H = \lambda \begin{bmatrix} \mathbf{I}_3 & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Rotation (3 parameters)

$$H = \lambda \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Rigid body transformation (6 parameters)

$$H = \lambda \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \quad \text{3 translation + 3 rotation}$$

Similarity transformation (7 parameters)

$$H = \lambda \begin{bmatrix} m\mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \quad \text{3 translation + 3 rotation + 1 scale angle preserving}$$

Affine transformation (12 parameters)

$$H = \lambda \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \quad \begin{array}{l} \text{3 translation + 3 rotation + 3 scale + 3 sheer} \\ \text{not angle preserving} \\ \text{parallel lines remain parallel} \end{array}$$

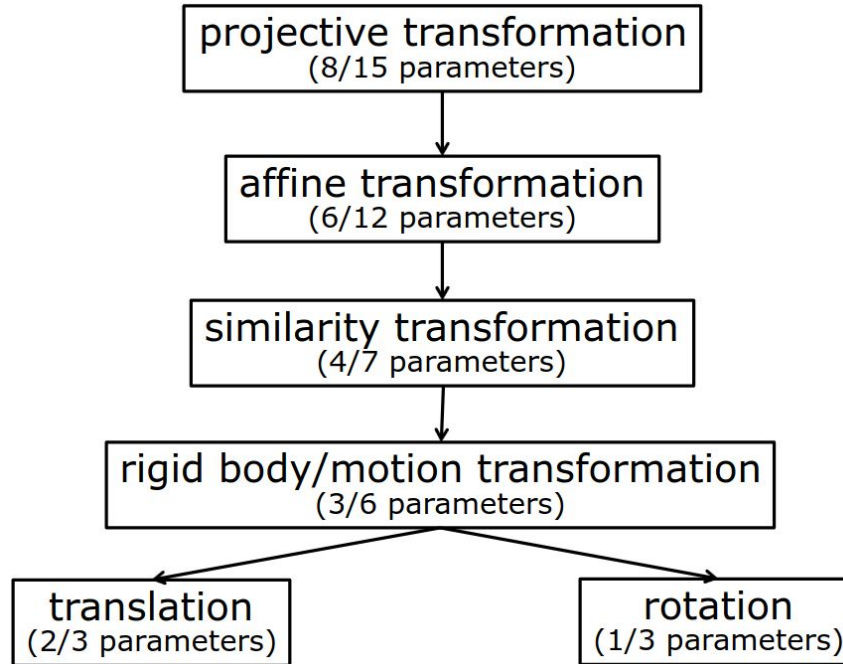
Projective transformation (15 parameters)

$$H = \lambda \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{a}^T & 1 \end{bmatrix} \quad \begin{array}{l} \text{affine transformation + 3 parameters} \\ \text{parallel lines may not stay parallel} \end{array}$$

2D Transformation	Figure	d. o. f.	H	H
Translation		2	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Mirroring at y-axis		1	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{Z} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Rotation		1	$\begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Motion		3	$\begin{bmatrix} \cos \varphi & -\sin \varphi & t_x \\ \sin \varphi & \cos \varphi & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Similarity		4	$\begin{bmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \lambda \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Scale difference		1	$\begin{bmatrix} 1+m/2 & 0 & 0 \\ 0 & 1-m/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Shear		1	$\begin{bmatrix} 1 & s/2 & 0 \\ s/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Asym. shear		1	$\begin{bmatrix} 1 & s' & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{S}' & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Affinity		6	$\begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$
Projectivity		8	$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$	$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{p}^T & 1/\lambda \end{bmatrix}$



Transformations Hierarchy



Inverting and Chaining

Inverting a transformation

$$\mathbf{X}' = H\mathbf{X}$$
$$\mathbf{X} = H^{-1}\mathbf{X}'$$

Chaining transformations via matrix products (not commutative)

$$\mathbf{X}' = H_1 H_2 \mathbf{X} \neq H_2 H_1 \mathbf{X}$$



Mapping the 3D World to an Image



$$\underset{\substack{\uparrow \\ \text{2D pixel coordinate}}}{\mathbf{x}} = \underset{\substack{\downarrow \\ \text{projection matrix}}}{P} \underset{\substack{\uparrow \\ \text{3D world coordinate}}}{\mathbf{X}}$$



Mapping the 3D World to an Image



projection matrix

$$\mathbf{x} = P\mathbf{X}$$

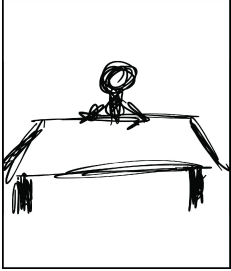
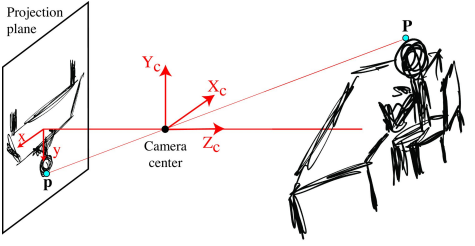
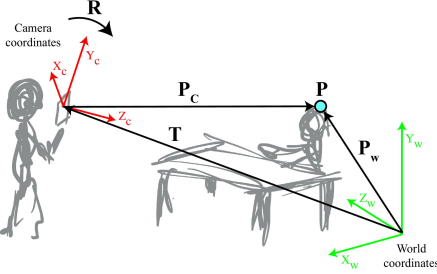
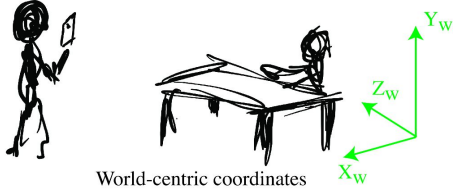
2D pixel coordinate 3D world coordinate

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = P_{3 \times 4} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

intrinsic & extrinsic

Mapping the 3D World to an Image $\mathbf{x} = P\mathbf{X}$

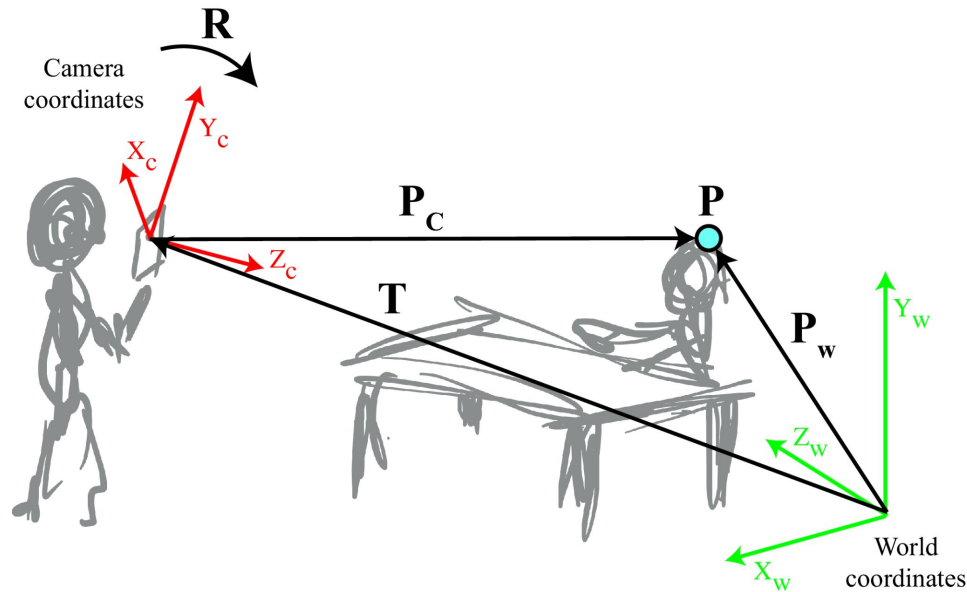
$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \mathbf{K}'\mathbf{T} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{X}_0 \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

			
Image plane (sensor system)	camera to image/sensor intrinsics	world to camera extrinsics	in the world system



Camera Extrinsics - Rigid Body Transformation

Extrinsics: Where is my camera in the 3D world?



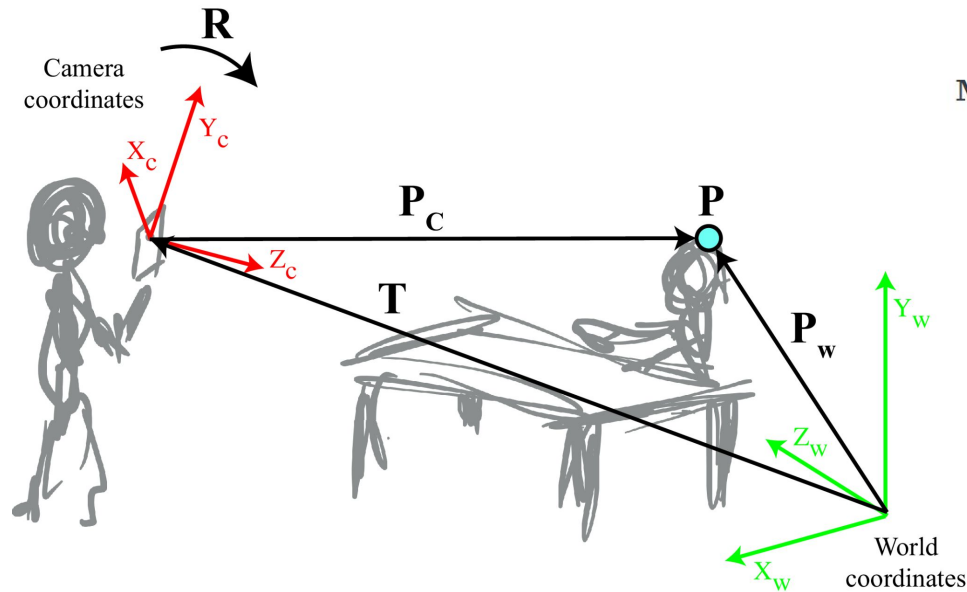
Heterogeneous coordinates

$$P_C = R(P_W - T) = RP_W - RT$$



Camera Extrinsics - Rigid Body Transformation

Extrinsics: Where is my camera in the 3D world?



Homogeneous coordinates

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 & 0 & -T_X \\ 0 & 1 & 0 & -T_Y \\ 0 & 0 & 1 & -T_Z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{M}_2 = \begin{bmatrix} R_1 & R_2 & R_3 & 0 \\ R_4 & R_5 & R_6 & 0 \\ R_7 & R_8 & R_9 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P}_C = \mathbf{M}_2 \mathbf{M}_1 \mathbf{P}_W$$

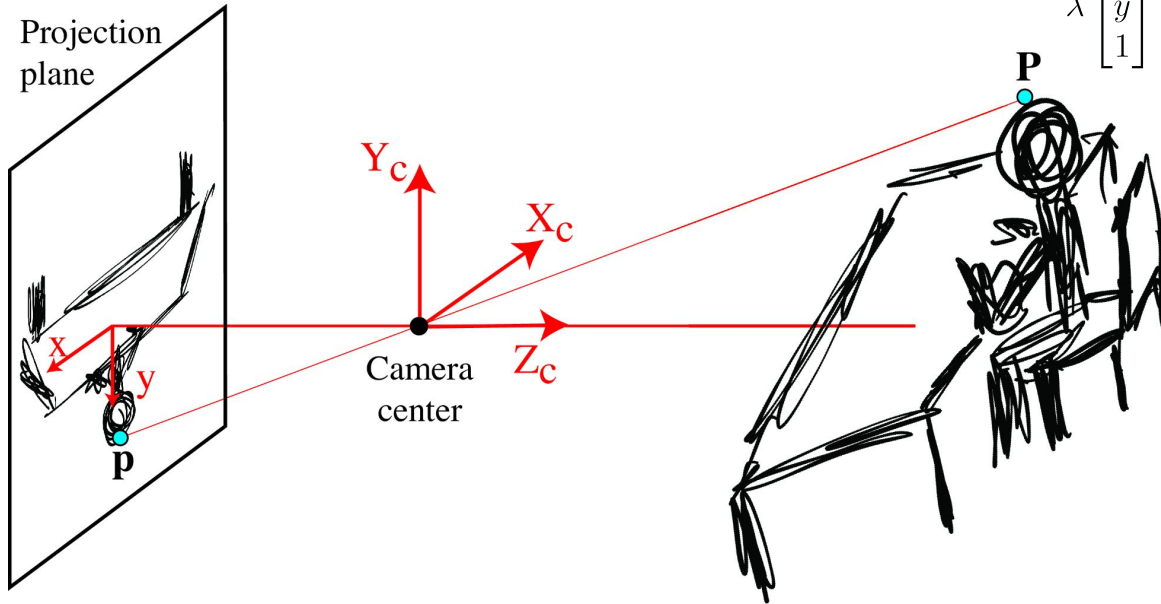
$$\mathbf{M}_2 \mathbf{M}_1 = \begin{bmatrix} R_1 & R_2 & R_3 & -\mathbf{r}_1 \mathbf{T} \\ R_4 & R_5 & R_6 & -\mathbf{r}_2 \mathbf{T} \\ R_7 & R_8 & R_9 & -\mathbf{r}_3 \mathbf{T} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & -\mathbf{R} \mathbf{T} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

$$\mathbf{P}_C = \begin{bmatrix} \mathbf{R} & -\mathbf{R} \mathbf{T} \\ \mathbf{0}^T & 1 \end{bmatrix} \mathbf{P}_W$$



Camera Intrinsics - Pinhole Model

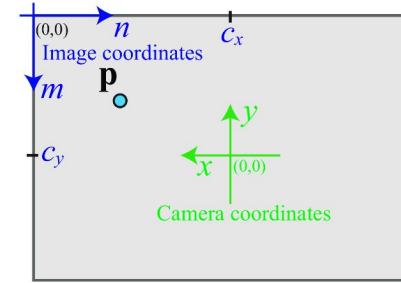
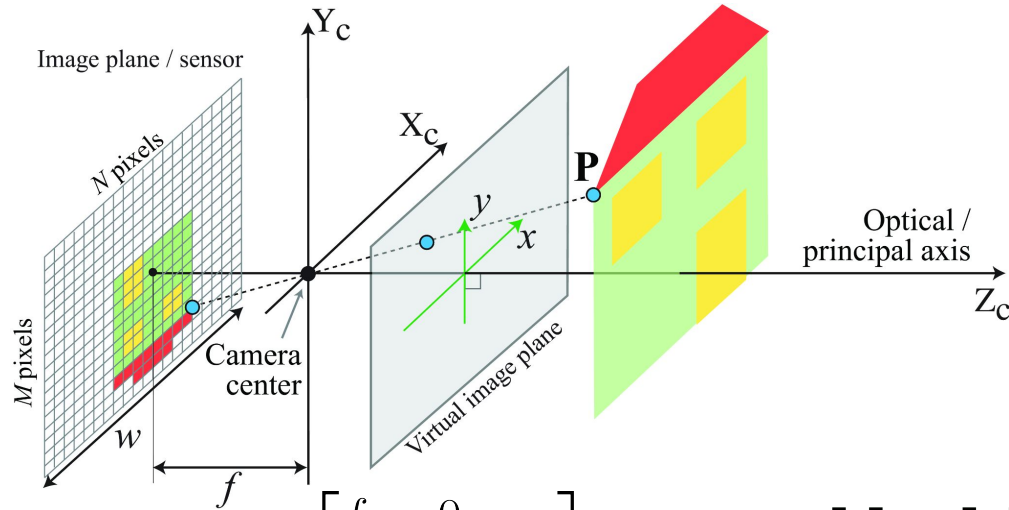
Intrinsics: Describe how to map a 3D world point on to the 2D image



$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \mathbf{K}' \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{P}_C$$



Camera Intrinsics - Pinhole Model



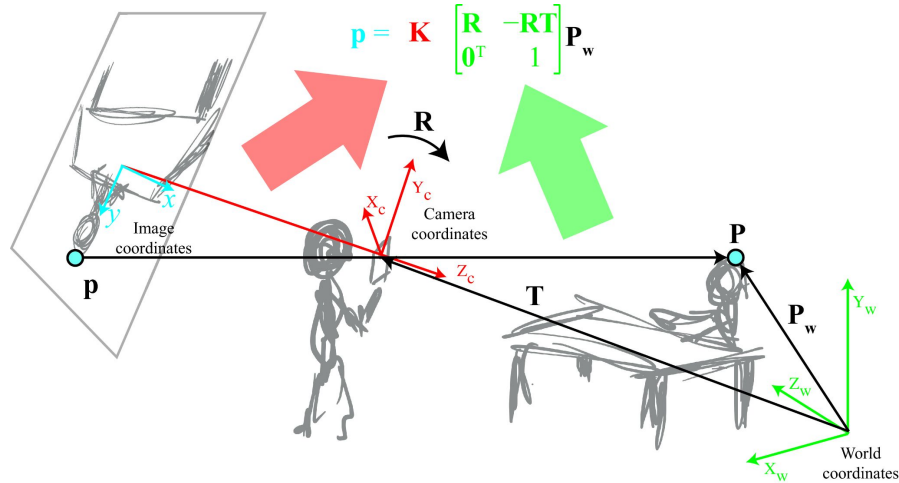
Convention 2
(c)

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = K' \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} P_C$$

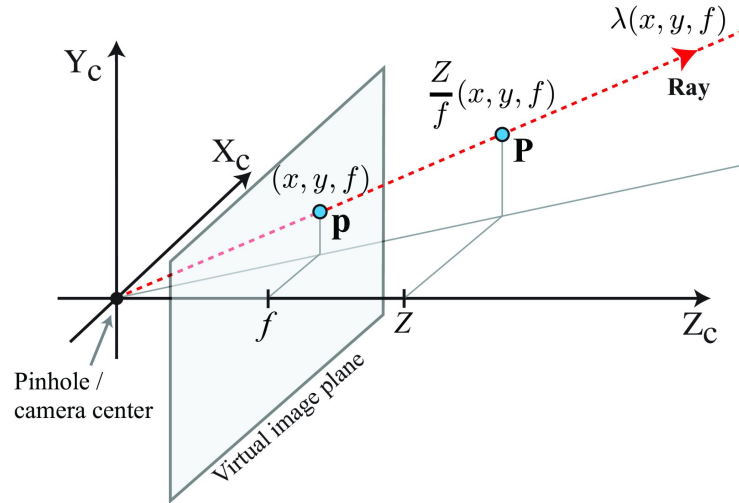
Mapping the 3D World to an Image

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \mathbf{K}'\mathbf{T} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{X}_0 \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



Inversion of the Mapping: From Pixels to Rays

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \mathbf{K}'\mathbf{T} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{X}_0 \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

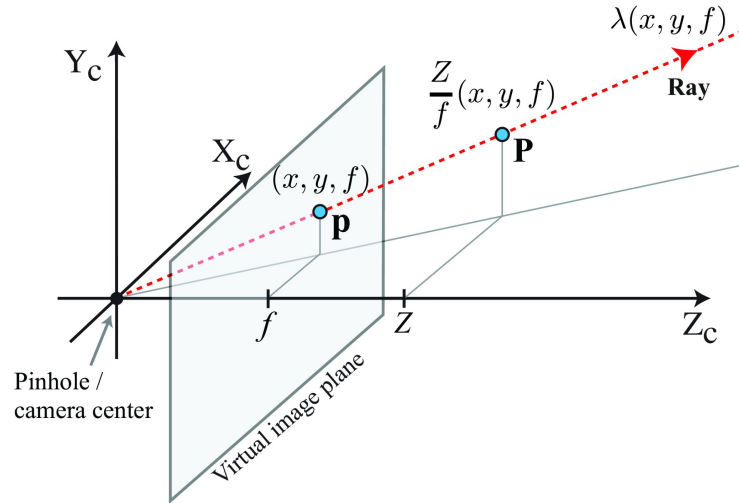


$$\lambda \mathbf{x} = \mathbf{K} \mathbf{R} \mathbf{X} - \mathbf{K} \mathbf{R} \mathbf{X}_0$$



Inversion of the Mapping: From Pixels to Rays

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \mathbf{K}'\mathbf{T} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{X}_0 \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

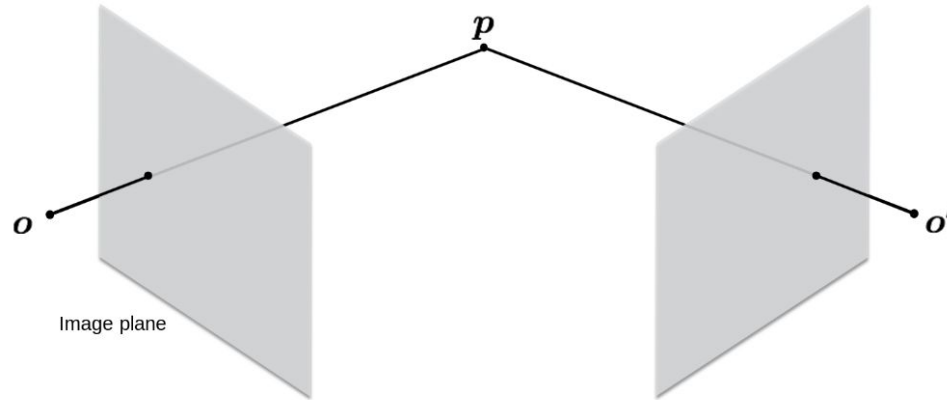


$$\lambda \mathbf{x} = \mathbf{K}\mathbf{R}\mathbf{X} - \mathbf{K}\mathbf{R}\mathbf{X}_0$$

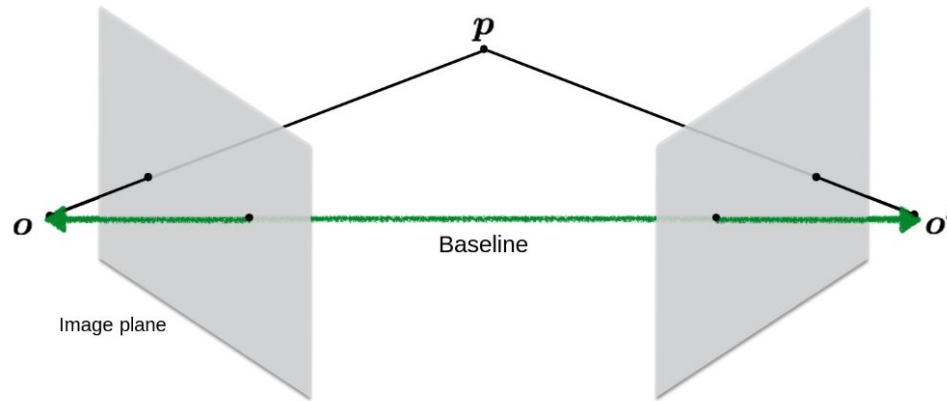
$$\mathbf{X} = \mathbf{X}_0 + \lambda(\mathbf{K}\mathbf{R})^{-1}\mathbf{x}$$



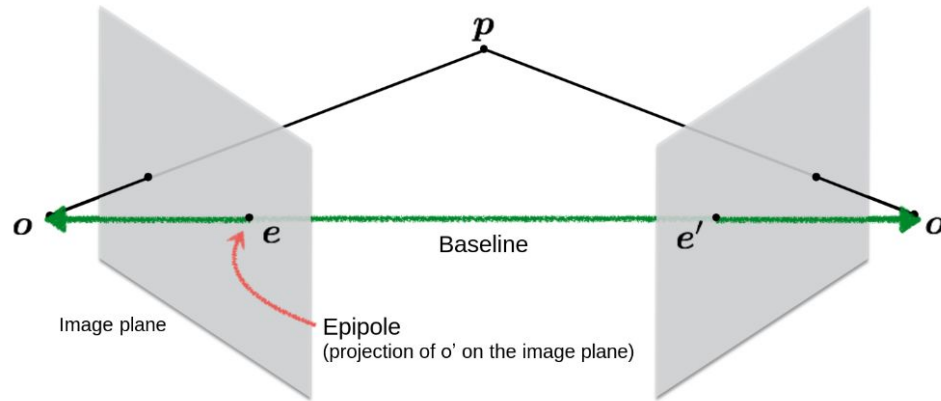
Two View Geometry: Notation



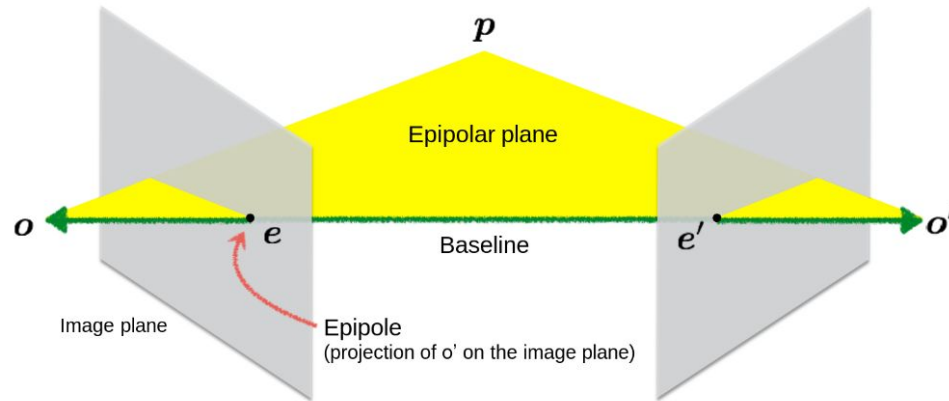
Two View Geometry: Notation



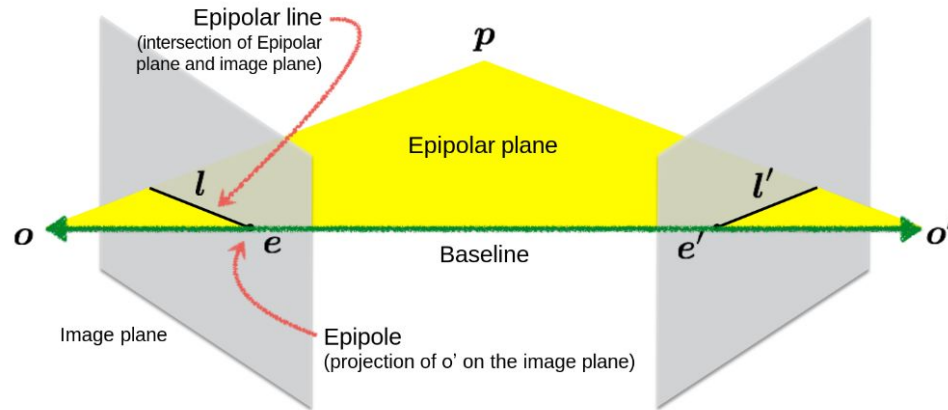
Two View Geometry: Notation



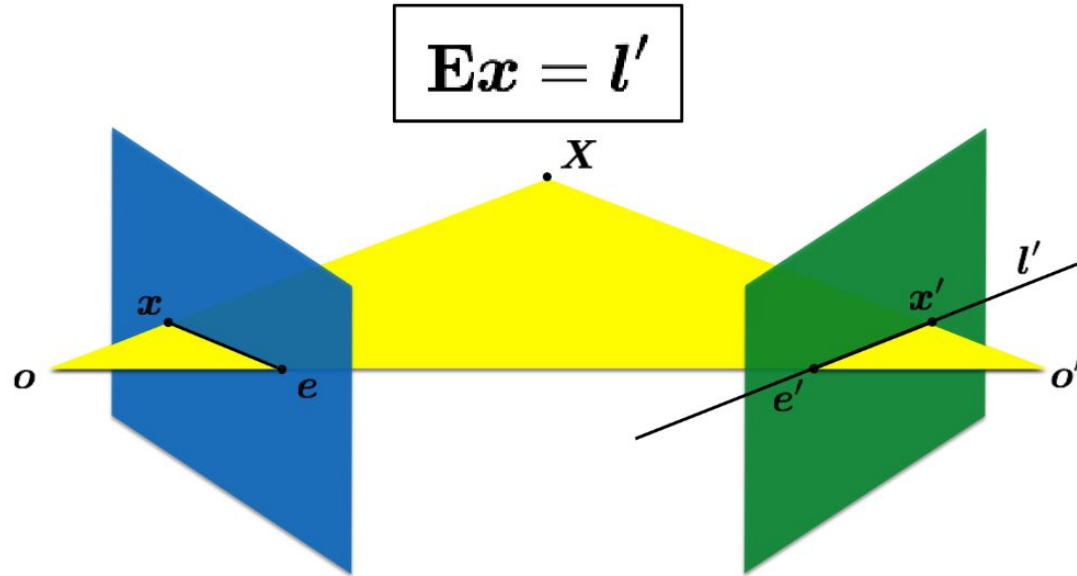
Two View Geometry: Notation



Two View Geometry: Notation



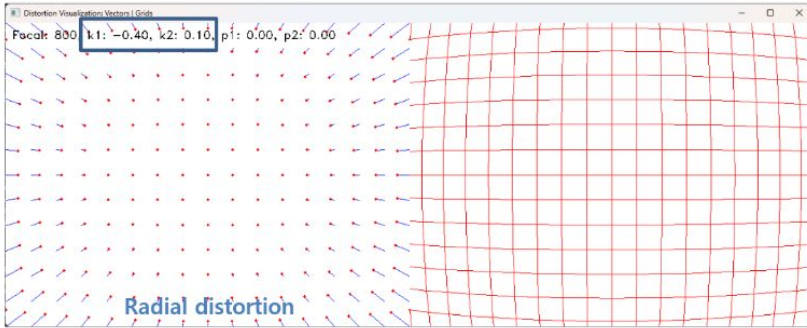
Two View Geometry: Essential Matrix



Camera Intrinsics - Distortion Model

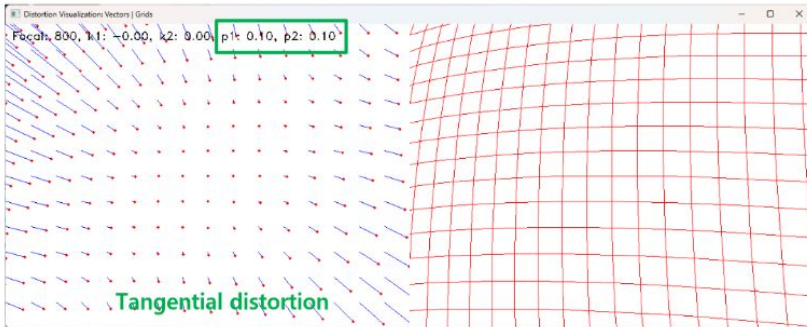


Camera Intrinsics - Distortion Model



Radial distortion

$$x_{distorted} = x(1 + k_1r^2 + k_2r^4 + k_3r^6) \quad y_{distorted} = y(1 + k_1r^2 + k_2r^4 + k_3r^6)$$



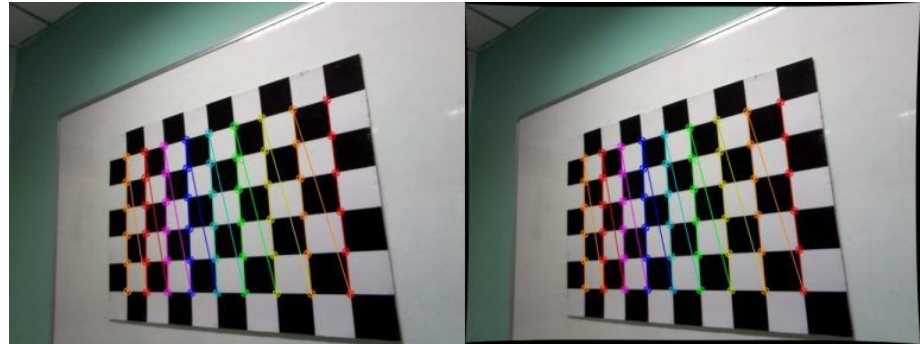
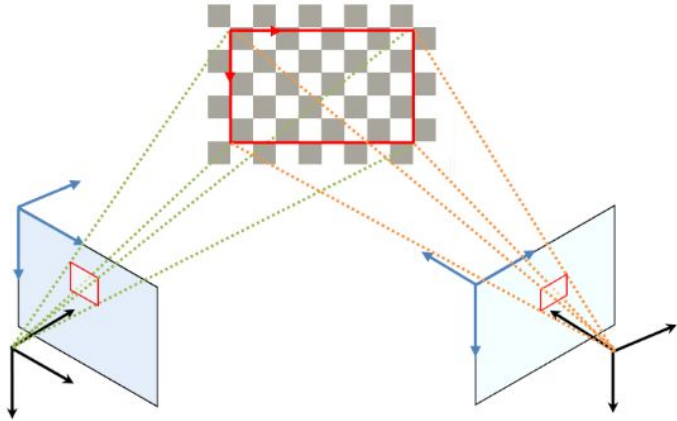
Tangential distortion

$$x_{distorted} = x + [2p_1xy + p_2(r^2 + 2x^2)] \quad y_{distorted} = y + [p_1(r^2 + 2y^2) + 2p_2xy]$$

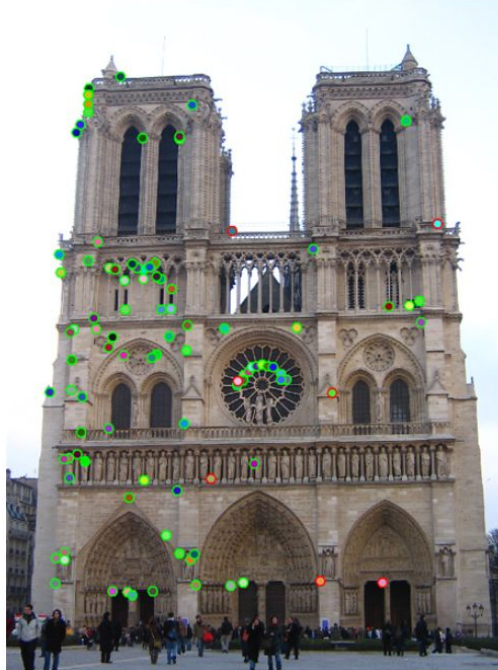
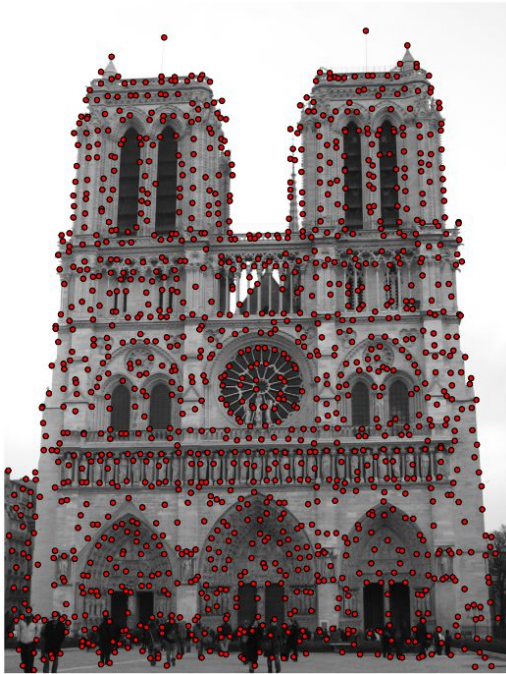
Camera Intrinsics - Distortion Model



Camera Intrinsics - Camera Calibration



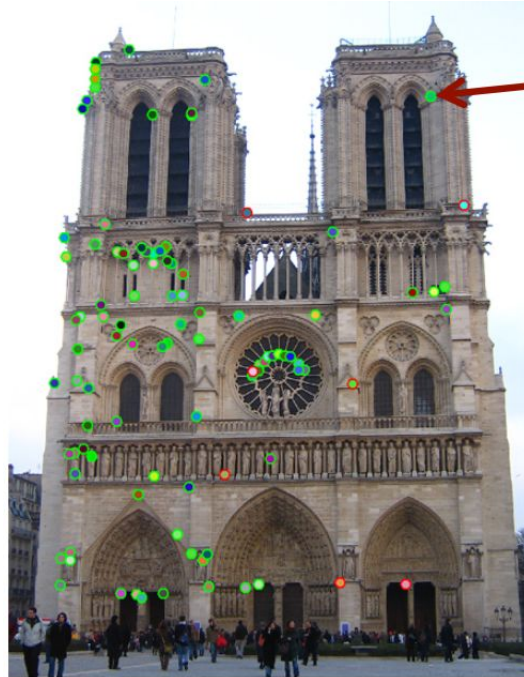
Visual Features: Keypoints & Descriptors



Visual Features: Keypoints & Descriptors

- ❖ **Keypoints:** Finding distinct points
 - **Harris** corners
 - **Shi-Tomasi**
 - **Förstner**
 - **Difference of Gaussians**

- ❖ **Features:** Describing a keypoint
 - **SIFT** - Scale Invariant Feature Transform
 - **BRIEF** - Binary Robust Independent Elementary Features
 - ❖ **ORB** - Oriented FAST Rotated BRIEF



keypoint

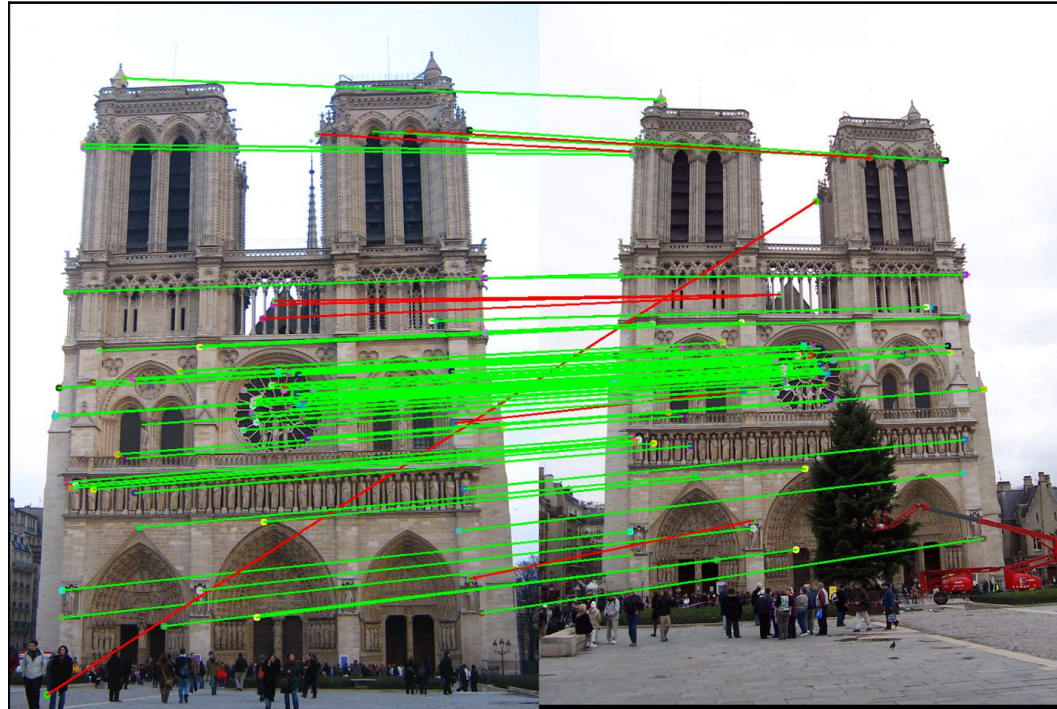
descriptor at
the keypoint

$$f = \begin{bmatrix} 0.02 \\ 0.04 \\ 0.1 \\ 0.03 \\ 0 \\ \dots \end{bmatrix}$$

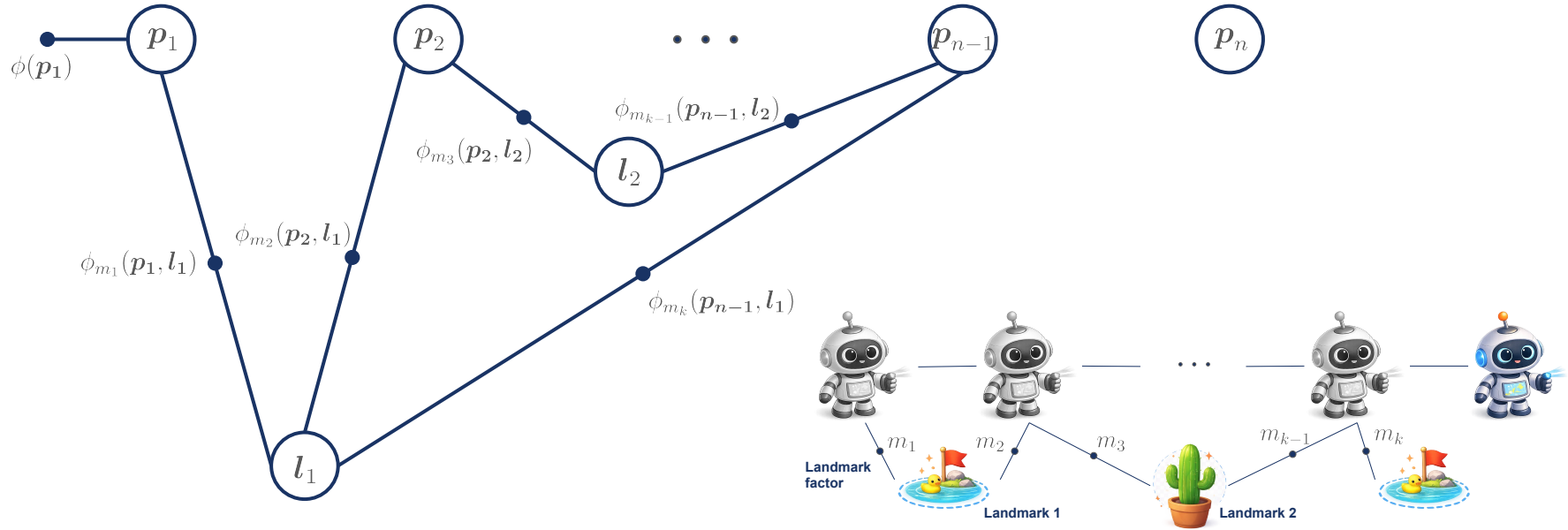


Visual Features: Matching

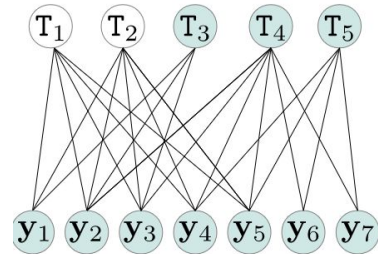
- Hamming Norm
- L2 Norm



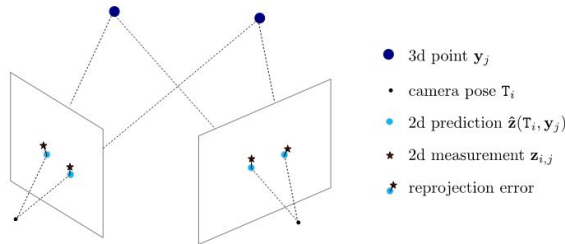
Bundle Adjustment - Landmark-based SLAM



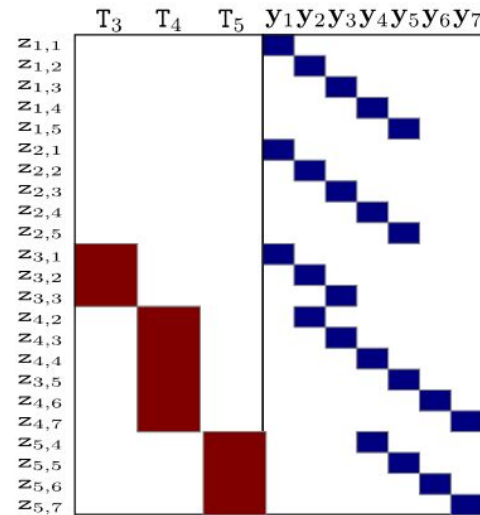
Bundle Adjustment - Landmark-based SLAM



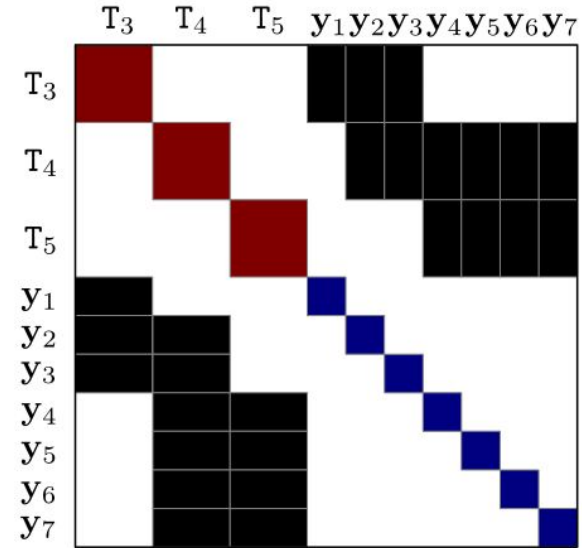
(a) Markov random field



- 3d point y_j
- camera pose T_i
- 2d prediction $\hat{z}(T_i, y_j)$
- 2d measurement $z_{i,j}$
- reprojection error



(b) Jacobian



(c) Hessian





Centre for
Robotics



SLAM Course

09 - Photogrammetry

From Projective Geometry to Bundle Adjustment

Dr Alejandro Fontán Villacampa
April 2026